

Toward Systematic Conveying of Architecture Design Knowledge for Self-Adaptive Systems

Sandro S. Andrade^{† §}

[§]GSORT Distributed Systems Group
Federal Institute of Education, Science
and Technology of Bahia, Salvador-Ba, Brazil
Email: sandros@ufba.br

Raimundo José de A. Macêdo[†]

[†]Distributed Systems Laboratory (LaSiD)
Department of Computer Science
Federal University of Bahia, Salvador-Ba, Brazil
Email: macedo@ufba.br

Abstract—With the increasing complexity and stringent requirements of modern large-scale distributed systems, well-structured representations of software design knowledge arise as a promising approach to keep delivering high quality products in a timely and cost-effective way. Although domain-specific architecture styles and reference architectures help in conveying such design knowledge, the lack of systematic and structured representations makes it hard to grasp design alternatives promptly and support well-informed trade-off analysis. This short paper presents DuSE-MT – a supporting tool for the DuSE approach to architectural design of self-adaptive systems. DuSE-MT implements: *i*) a generic metamodel for systematic representation of design spaces (DuSE), which enables automated architecture design and analysis; *ii*) a specific design space for the self-adaptive systems domain (SA:DuSE); *iii*) a set of metrics that capture quality attributes of resulting self-adaptive architectures; and *iv*) a multi-objective optimization approach to explicitly elicit a trade-off decision by finding out a set of Pareto-optimal candidate architectures. Our approach has been evaluated in a case study involving self-adaptive cloud-based services.

I. INTRODUCTION

Designing architectures for software-intensive systems that fulfill complex quality attributes and operate in highly unpredictable environments is a challenging task [1]. The well-orchestrated use of refined experience, good aesthetics, and right balance when making trade-off decisions is imperative if we are to design effective architectures for complex large-scale systems. Furthermore, developing systems for specific application domains ultimately yields a body of tacit design knowledge that would be of paramount value if systematically structured for routine reuse.

In particular, the design of effective architectures for self-adaptive systems requires architects become proficient in evaluating complex solution spaces [1]. Relevant approaches scattered over different research communities and solutions effective only in specific scenarios make it harder to understand generic architecture-centric approaches to self-adaptation.

In this paper, we present DuSE-MT¹ – a supporting tool for the DuSE approach to automated architectural design and analysis of self-adaptive systems, presented in [2]. DuSE-MT fully implements: *i*) a generic metamodel for systematic representation of design spaces (DuSE), which enables automated

architecture design and analysis; *ii*) a specific design space for the self-adaptive systems domain (SA:DuSE); *iii*) a set of metrics that capture quality attributes of resulting self-adaptive architectures; and *iv*) a multi-objective optimization approach to explicitly elicit a trade-off decision by finding out a set of Pareto-optimal candidate architectures.

A. Challenges when Designing Self-Adaptive Systems

Although a lot of current research have been devoted to the use of feedback control theory as the enabling mechanism for self-adaptation, it seems hard – even for experienced architects – to understand and evaluate trade-offs among currently available approaches. The variety of solutions available in many design dimensions rapidly opens up a huge space of candidate architectures for self-adaptive behavior. In particular, the following challenges are frequently faced when designing self-adaptive systems:

C1) Highly specialized domain and large design/solution space: deciding about a specific system modeling approach, control law, tuning method, and arrangements of feedback loops, heavily impacts not only system's self-adaptation robustness and control overhead, but also common quality attributes such as scalability, flexibility and maintainability. Currently, self-adaptation is a mean to meet several different goals and a number of mechanisms for feedback control are available. It is very likely that novice architects adopt inferior architectures or be unaware of alternative solutions.

C2) Lack of support for early reasoning of quality properties: as self-adaptation capabilities become essential means to meet business goals, it is highly desirable that final product's quality attributes can be evaluated/estimated at early development stages. Since architectural choices directly impact such attributes, it is reasonable that systematic representations of design knowledge for self-adaptive systems also include mechanisms for early quality attributes assessment. Such mechanisms set the stage for well-informed trade-off decisions and help architects to easily grasp the consequences of choosing a particular candidate architecture.

C3) Lack of effective generic solutions: although a number of noteworthy mechanisms for self-adaptation have been proposed over the past years, they are usually tailored to very specific scenarios, which make it harder to reuse their solutions in other applications. Whilst optimization and configuration for the problem at hand are commonly required,

This research is partially supported by grant 006/2012/PRPGI from the IFBa's Research and Innovation Supporting Program.

¹<http://duse.sf.net>

